

Statistical Appendix

Singapore Birth and Fertility Time Series — MATH X313

Zhu Ying (a1984460)

today

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```
library(readr)
library(dplyr)
library(knitr)

table_dir <- file.path("../", "outputs", "tables")
figure_dir <- file.path("../", "outputs", "figures")
```

1 Data Preparation Details

The analysis uses `data/raw/BirthsAndFertilityRatesAnnual.csv`, obtained from the Singapore Department of Statistics (SingStat) births and fertility register. The raw file is in wide format: each row is a named demographic series and each column is a calendar year. The preparation script `scripts/01_prepare_data.R` performs the following steps:

1. Reads the raw CSV and identifies year columns by pattern matching.
2. Pivots to long format with columns `Measure`, `Year`, and `Observed`.
3. Parses year columns as integers and observed values as numeric.

4. Filters to the two analysis series (Total Fertility Rate and Total Live Births) by exact name match after trimming whitespace.
5. Constructs a `tsibble` time series object indexed by `Measure` and `Year`.

The validation script `scripts/01_validate_data.R` then confirms the cleaned data meet all required quality checks before any analysis proceeds.

```
clean_series <- read_csv(file.path(table_dir, "clean_birth_fertility_series.csv"),
                        show_col_types = FALSE)
kable(head(clean_series, 10), digits = 4)
```

Table 1: First ten rows of the cleaned analysis-ready series

Measure	Year	Observed
Total fertility rate	1960	5.76
Total fertility rate	1961	5.41
Total fertility rate	1962	5.21
Total fertility rate	1963	5.16
Total fertility rate	1964	4.97
Total fertility rate	1965	4.66
Total fertility rate	1966	4.46
Total fertility rate	1967	3.91
Total fertility rate	1968	3.53
Total fertility rate	1969	3.22
Total fertility rate	1970	3.07
Total fertility rate	1971	3.02
Total fertility rate	1972	3.04
Total fertility rate	1973	2.79
Total fertility rate	1974	2.35
Total fertility rate	1975	2.07
Total fertility rate	1976	2.11
Total fertility rate	1977	1.82
Total fertility rate	1978	1.79
Total fertility rate	1979	1.79
Total fertility rate	1980	1.82
Total fertility rate	1981	1.78
Total fertility rate	1982	1.74
Total fertility rate	1983	1.61
Total fertility rate	1984	1.62
Total fertility rate	1985	1.61
Total fertility rate	1986	1.43
Total fertility rate	1987	1.62
Total fertility rate	1988	1.96
Total fertility rate	1989	1.75
Total fertility rate	1990	1.83
Total fertility rate	1991	1.73
Total fertility rate	1992	1.72
Total fertility rate	1993	1.74
Total fertility rate	1994	1.71
Total fertility rate	1995	1.67
Total fertility rate	1996	1.66
Total fertility rate	1997	1.61

Measure	Year	Observed
Total fertility rate	1998	1.48
Total fertility rate	1999	1.47
Total fertility rate	2000	1.6
Total fertility rate	2001	1.41
Total fertility rate	2002	1.37
Total fertility rate	2003	1.27
Total fertility rate	2004	1.26
Total fertility rate	2005	1.26
Total fertility rate	2006	1.28
Total fertility rate	2007	1.29
Total fertility rate	2008	1.28
Total fertility rate	2009	1.22
Total fertility rate	2010	1.15
Total fertility rate	2011	1.2
Total fertility rate	2012	1.29
Total fertility rate	2013	1.19
Total fertility rate	2014	1.25
Total fertility rate	2015	1.24
Total fertility rate	2016	1.2
Total fertility rate	2017	1.16
Total fertility rate	2018	1.14
Total fertility rate	2019	1.14
Total fertility rate	2020	1.1
Total fertility rate	2021	1.12
Total fertility rate	2022	1.04
Total fertility rate	2023	0.97
Total fertility rate	2024	0.97
Total fertility rate	2025	0.87
Total live births	1960	61775
Total live births	1961	59930
Total live births	1962	58977
Total live births	1963	59530
Total live births	1964	58217
Total live births	1965	55725
Total live births	1966	54680
Total live births	1967	50560
Total live births	1968	47241
Total live births	1969	44562
Total live births	1970	45934
Total live births	1971	47088
Total live births	1972	49678
Total live births	1973	48269
Total live births	1974	43268
Total live births	1975	39948
Total live births	1976	42783
Total live births	1977	38364
Total live births	1978	39441
Total live births	1979	40779
Total live births	1980	41217
Total live births	1981	42250

Measure	Year	Observed
Total live births	1982	42654
Total live births	1983	40585
Total live births	1984	41556
Total live births	1985	42484
Total live births	1986	38379
Total live births	1987	43616
Total live births	1988	52957
Total live births	1989	47669
Total live births	1990	51142
Total live births	1991	49114
Total live births	1992	49402
Total live births	1993	50225
Total live births	1994	49554
Total live births	1995	48635
Total live births	1996	48577
Total live births	1997	47333
Total live births	1998	43664
Total live births	1999	43336
Total live births	2000	46997
Total live births	2001	41451
Total live births	2002	40760
Total live births	2003	37485
Total live births	2004	37174
Total live births	2005	37492
Total live births	2006	38317
Total live births	2007	39490
Total live births	2008	39826
Total live births	2009	39570
Total live births	2010	37967
Total live births	2011	39654
Total live births	2012	42663
Total live births	2013	39720
Total live births	2014	42232
Total live births	2015	42185
Total live births	2016	41251
Total live births	2017	39615
Total live births	2018	39039
Total live births	2019	39279
Total live births	2020	38590
Total live births	2021	38672
Total live births	2022	35605
Total live births	2023	33541
Total live births	2024	33703
Total live births	2025	30004

```
kable(tail(clean_series, 10), digits = 4)
```

```
series_summary <- read_csv(file.path(table_dir, "series_summary.csv"),
                           show_col_types = FALSE)
```

```
kable(series_summary, digits = 2)
```

Table 2: Series summary

Measure	first_year	last_year	observations	first_observation	latest_observation
Total fertility rate	1960	2025	66	5.76	0.87
Total live births	1960	2025	66	61775	30004

```
validation_checks <- read_csv(file.path(table_dir, "data_validation_checks.csv"),
                               show_col_types = FALSE)
kable(validation_checks)
```

Table 3: Automated data validation checks

check	passed	detail
expected_measures_present	TRUE	Total fertility rate; Total live births
no_missing_values	TRUE	0 missing observations
no_duplicate_measure_year_keys	TRUE	0 duplicate keys
annual_years_are_consecutive	TRUE	0 series with non-consecutive years
positive_observations	TRUE	0 non-positive observations
minimum_history_length	TRUE	minimum observations per measure: 66

All six validation checks passed with no corrections required. The analysis window is 1960–2025, yielding 66 annual observations per series. No imputation or interpolation was performed.

2 Additional Exploratory Plots

Level ACF

```
knitr::include_graphics(file.path(figure_dir, "level_acf.png"))
```

The level ACFs (@fig-level-acf) decay very slowly and monotonically, with virtually every lag exceeding the 95% confidence bands. This pattern is characteristic of an integrated (I(1)) or near-unit-root process. Modelling at the level without differencing would leave strong autocorrelation in the residuals, violating the white-noise assumption of ARIMA models.

First-Difference ACF

```
knitr::include_graphics(file.path(figure_dir, "first_difference_acf.png"))
```

After first differencing (@fig-diff-acf):

- **TFR:** A positive spike at lag 1 (ACF1 = 0.37) remains within the confidence band boundary. This single significant lag motivates the MA(1) term in the ARIMA(0,1,1) selection.
- **Total Live Births:** No spikes beyond lag 0 are significant, consistent with the ARIMA(0,1,0) (random walk) selection. The differenced series behaves as approximately white noise.

Autocorrelation in the original series

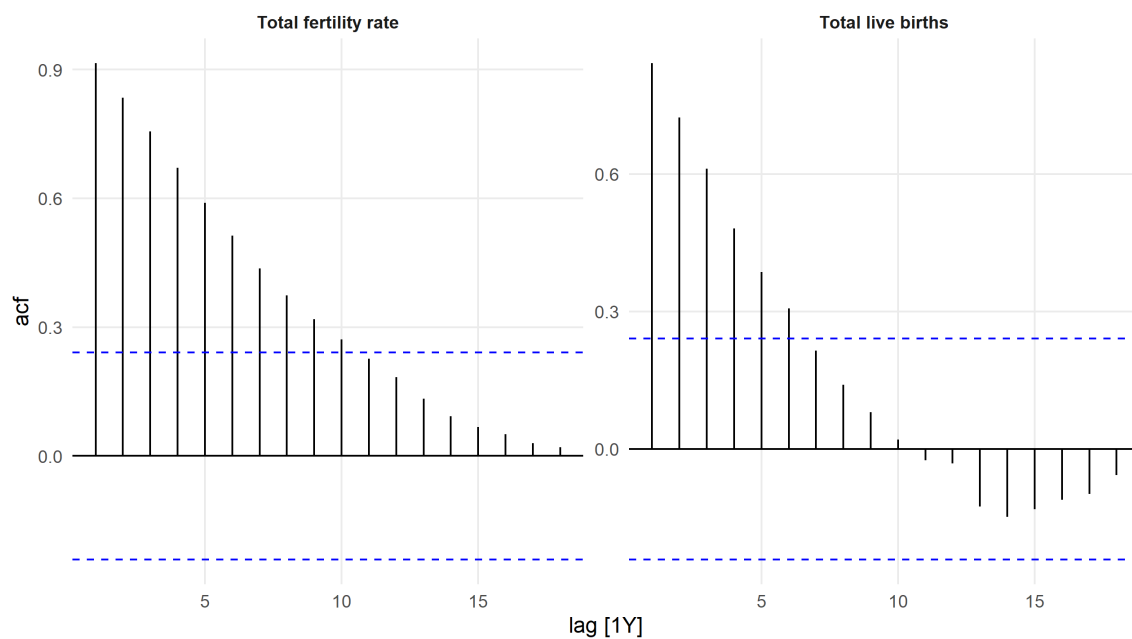


Figure 1: Sample ACF of both series at the original (level) scale.

Autocorrelation after first differencing

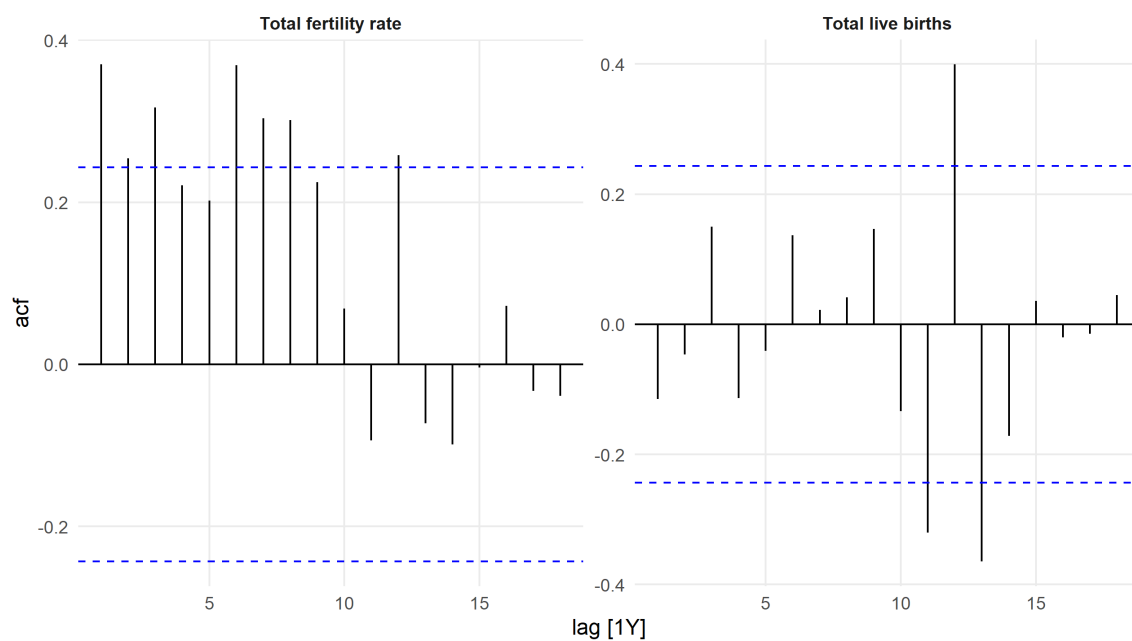


Figure 2: Sample ACF after one order of first differencing.

3 Model Equations

Benchmark Models

Naïve model:

$$\hat{y}_{T+h|T} = y_T \quad \text{for all } h \geq 1$$

The point forecast is always the last observed value. Prediction intervals widen at rate $\sqrt{h}$.

Drift model (random walk with drift):

$$y_t = y_{t-1} + \hat{b} + \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

where $\hat{b}$ is estimated as the average of all historical first differences:

$$\hat{b} = \frac{y_T - y_1}{T - 1}$$

For TFR: $\hat{b} = -0.082$ (SE = 0.021). For Total Live Births: $\hat{b} = -356$ births per year (SE = 371).

Deterministic Trend Model

$$y_t = \beta_0 + \beta_1 \cdot t + \varepsilon_t, \quad \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

For TFR: $\hat{\beta}_0 = 4.056$, $\hat{\beta}_1 = -0.0645$ (both $p < 0.001$). For Total Live Births: $\hat{\beta}_0 = 52,542$, $\hat{\beta}_1 = -249.5$ (both $p < 0.001$). Despite highly significant coefficients, the residuals of this model carry strong autocorrelation for both series (see @sec-diagnostics), invalidating the white-noise assumption.

ARIMA Models

The general ARIMA(p, d, q) model for a time series y_t after d orders of differencing is:

$$\phi(B)(1 - B)^d y_t = \theta(B) \varepsilon_t$$

where $\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$ is the AR polynomial, $\theta(B) = 1 - \theta_1 B - \dots - \theta_q B^q$ is the MA polynomial, B is the backshift operator, and $\varepsilon_t \sim \text{WN}(0, \sigma^2)$.

TFR — ARIMA(0,1,1): $p = 0, d = 1, q = 1$:

$$(1 - B) y_t = (1 + \theta_1 B) \varepsilon_t$$

$$y_t - y_{t-1} = \varepsilon_t - 0.799 \varepsilon_{t-1}$$

Estimated innovation variance: $\hat{\sigma}^2 = 0.0196$. AICc = -54.80. This model is equivalent to simple exponential smoothing (SES) on the differenced series with smoothing parameter $\alpha = 1 - |\theta_1| = 0.201$.

Total Live Births — ARIMA(0,1,0):

$$(1 - B) y_t = \varepsilon_t \implies y_t - y_{t-1} = \varepsilon_t$$

Estimated innovation variance: $\hat{\sigma}^2 = 7,542,931$. AICc = 1029.14. This is a pure random walk; the Drift model (above) is preferred for live births because it captures the persistent declining trend.

ETS Model

The ETS model selected by the fable package is of the form ETS(A, Ad, N) for TFR — additive error, damped additive trend, no seasonality:

$$\begin{aligned}y_t &= \ell_{t-1} + \phi b_{t-1} + \varepsilon_t \\ \ell_t &= \ell_{t-1} + \phi b_{t-1} + \alpha \varepsilon_t \\ b_t &= \phi b_{t-1} + \beta \varepsilon_t\end{aligned}$$

Estimated parameters for TFR: $\alpha = 0.993$, $\beta = 0.0001$, $\phi = 0.901$, $\ell_0 = 6.199$, $b_0 = -0.511$. The near-unity α implies very fast adaptation of the level, while the small β and damped $\phi < 1$ produce a slowly decaying trend component.

For Total Live Births the selected ETS is ETS(A, N, N) — simple exponential smoothing with additive error and no trend or seasonal component:

$$y_t = \ell_{t-1} + \varepsilon_t, \quad \ell_t = \ell_{t-1} + \alpha \varepsilon_t$$

Estimated: $\alpha = 0.890$, $\ell_0 = 61,570$.

4 Model Selection Tables

```
fit_stats <- read_csv(file.path(table_dir, "candidate_model_fit_statistics.csv"),
                      show_col_types = FALSE) |>
  select(Measure, .model, AIC, AICc, BIC)
kable(fit_stats, digits = 2)
```

Table 4: Candidate model fit statistics (training sample, 1960–2015)

Measure	.model	AIC	AICc	BIC
Total fertility rate	naive			
Total fertility rate	drift			
Total fertility rate	linear_trend	-35.0939	-34.6324	-29.0179
Total fertility rate	auto_arima	-55.0317	-54.7964	-51.0537
Total fertility rate	ets	4.27853	5.99282	16.4306
Total live births	naive			
Total live births	drift			
Total live births	linear_trend	959.09	959.551	965.166
Total live births	auto_arima	1029.07	1029.14	1031.08
Total live births	ets	1116.57	1117.03	1122.64

```
ljung_box <- read_csv(file.path(table_dir, "residual_ljung_box_tests.csv"),
                      show_col_types = FALSE)
kable(ljung_box, digits = 4,
      col.names = c("Series", "Model", "LB statistic", "p-value"))
```


Table 5: Ljung-Box residual test (lag = 10)

Series	Model	LB statistic	p-value
Total fertility rate	auto_arima	11.1487	0.34605
Total fertility rate	drift	44.5183	2.65594e-06
Total fertility rate	ets	6.88353	0.736393
Total fertility rate	linear_trend	177.473	0
Total fertility rate	naive	44.5183	2.65594e-06
Total live births	auto_arima	7.64524	0.663444
Total live births	drift	7.58541	0.669262
Total live births	ets	6.54603	0.767494
Total live births	linear_trend	109.597	0
Total live births	naive	7.58541	0.669262

```
knitr::include_graphics(file.path(ffigure_dir, "candidate_model_residuals.png"))
```

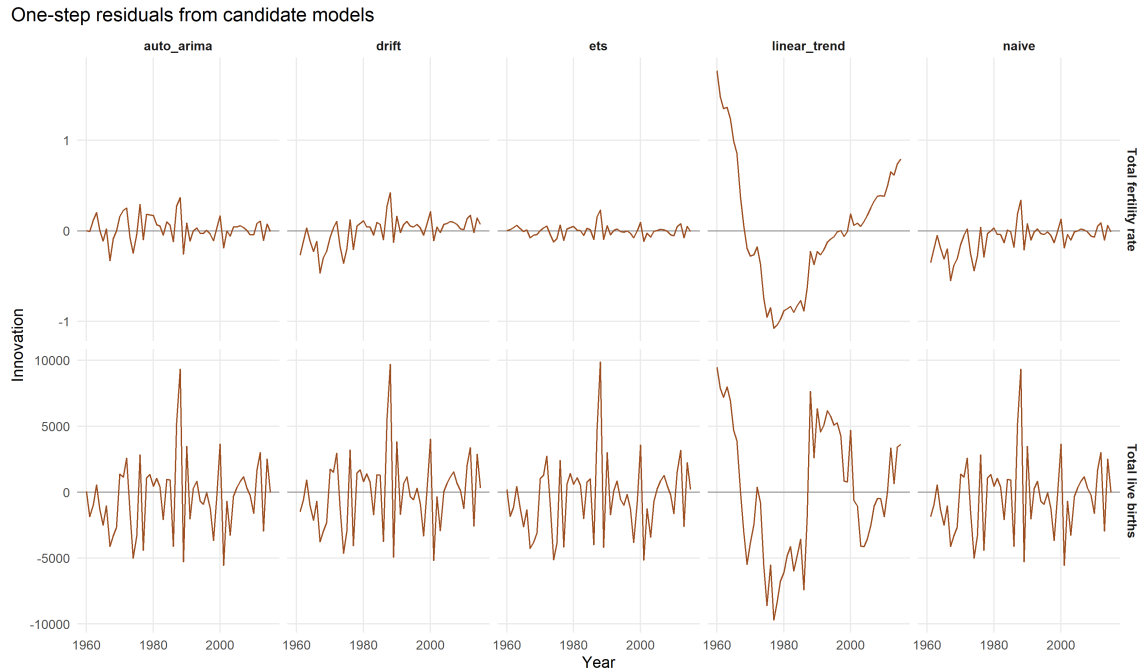


Figure 3: One-step training residuals by model and series.

```
knitr::include_graphics(file.path(ffigure_dir, "candidate_residual_acf.png"))

selection_evidence <- read_csv(file.path(table_dir, "model_selection_evidence.csv"),
                               show_col_types = FALSE) |>
  select(Measure, .model, RMSE, MAE, MAPE, AICc,
         ljung_box_p_value, residual_pass_ljung_box_0_05, review_flag)
kable(selection_evidence, digits = 3)
```

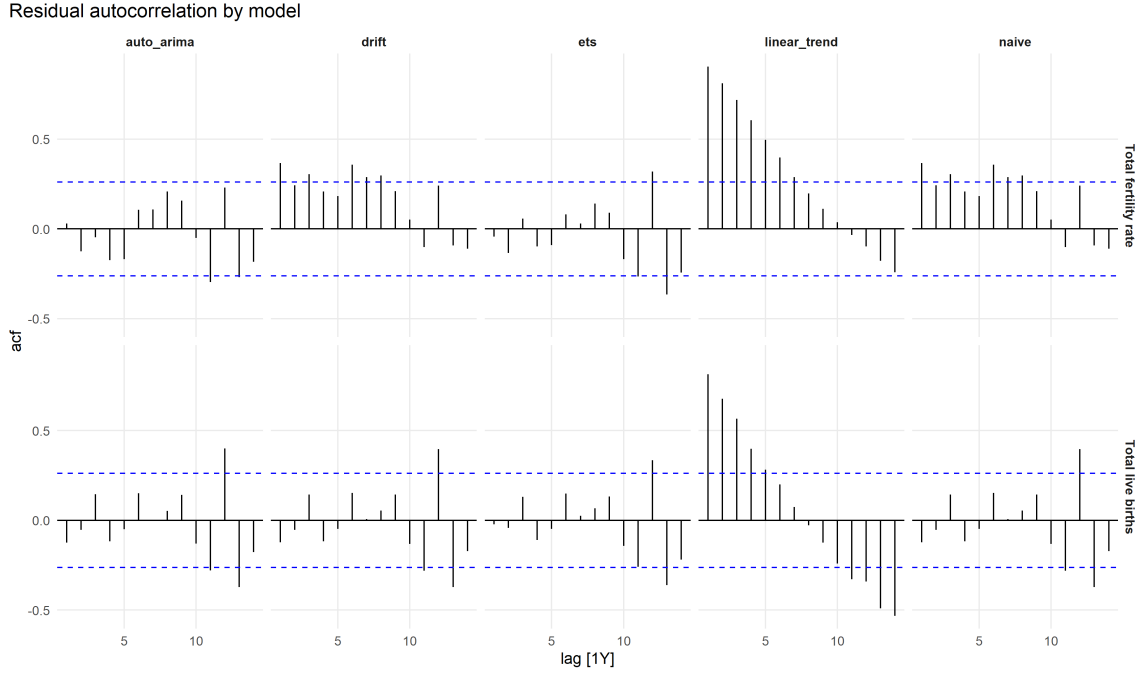


Figure 4: Residual ACF (lags 1–14) for each candidate model.

Table 6: Combined model selection evidence

.model	Measure	RMSE	MAE	MAPE	AICc	ljung_box_residual_pass_ljung_box_flag_05	
auto_arima	Total fertility rate	0.178424	0.153068	8.2901	-	0.34605	TRUE
				54.7964			accuracy_and_residuals
ets	Total fertility rate	0.190390	0.163577	6.334	5.99282	0.736393	TRUE
							secondary_candidate
naive	Total fertility rate	0.196138	0.169	16.8645		2.65594e-06	FALSE
							lower_priority
drift	Total fertility rate	0.317950	0.283	27.8023		2.65594e-06	FALSE
							lower_priority
linear_trend	Total fertility rate	0.984110	0.979439	92.9922	-	0	FALSE
				34.6324			lower_priority
linear_trend	Total live births	2718.992321	326.636759	59.551		0	FALSE
							accuracy_only
drift	Total live births	4097.4	3296.1	9.65776		0.669262	TRUE
							secondary_candidate
ets	Total live births	6221.955233	2215.202811	17.03	0.767494		TRUE
							lower_priority
auto_arima	Total live births	6240.365255	15.262510	29.14	0.663444		TRUE
							lower_priority
naive	Total live births	6240.365255	15.2625		0.669262		TRUE
							lower_priority

5 Models Not Selected

The table below records the reason each non-selected candidate was rejected.

Series	Rejected model	Main reason	Supporting evidence
TFR	Naïve	Fails Ljung-Box ($p < 0.001$); higher RMSE than ARIMA	@tbl-ljung-box; @tbl-holdout-accuracy
TFR	Drift	Fails Ljung-Box ($p < 0.001$); RMSE = 0.318 vs ARIMA 0.178	@tbl-ljung-box; @tbl-holdout-accuracy
TFR	Linear trend	Fails Ljung-Box (LB = 177.5, $p < 0.001$); MAPE = 93%	@tbl-ljung-box; @fig-resid-acf
TFR	ETS	Passes diagnostics but higher RMSE (0.190) than ARIMA (0.178)	@tbl-holdout-accuracy
Live Births	Naïve	Identical to ARIMA(0,1,0); lower RMSE than Drift	—
Live Births	ARIMA(0,1,0)	Equivalent to Naïve; Drift captures historical trend better	@tbl-holdout-accuracy
Live Births	ETS	Passes diagnostics but RMSE = 6,222 vs Drift 4,097	@tbl-holdout-accuracy
Live Births	Linear trend	Fails Ljung-Box (LB = 109.6, $p < 0.001$); invalid PIs	@tbl-ljung-box

The linear trend model deserves specific comment for Total Live Births. Its holdout RMSE (2,719) is the lowest of all five candidates, which might suggest it is the best model by a simple accuracy criterion. However, a model with residual autocorrelation this severe (LB = 109.6) cannot produce valid prediction intervals: its stated uncertainty will be systematically too narrow (because autocorrelated residuals carry information that has not been extracted by the model). For a forecast whose primary value is in quantifying uncertainty as well as direction, this disqualifies the linear trend model from selection.

6 Holdout Forecast Evaluation

```
holdout_accuracy <- read_csv(file.path(table_dir, "holdout_forecast_accuracy.csv"),
                             show_col_types = FALSE) |>
  select(Measure, .model, ME, RMSE, MAE, MAPE, ACF1) |>
  arrange(Measure, RMSE)
kable(holdout_accuracy, digits = 3)
```

Table 8: Holdout forecast accuracy (2016–2025, all candidates)

.model	Measure	ME	RMSE	MAE	MAPE	ACF1
auto_arima	Total fertility rate	-0.153068	0.178424	0.153068	15.2901	0.570258
ets	Total fertility rate	-0.163577	0.190391	0.163577	16.334	0.580918
naive	Total fertility rate	-0.169	0.196138	0.169	16.8645	0.584509
drift	Total fertility rate	0.283	0.317951	0.283	27.8023	0.718291
linear_trend	Total fertility rate	0.979432	0.984112	0.979432	92.9922	0.705948
linear_trend	Total live births	-265.685	2718.99	2321.32	6.63675	0.523552
drift	Total live births	-3296.1	4097.4	3296.1	9.65776	0.495506
ets	Total live births	-5233.22	6221.95	5233.22	15.2028	0.569298
auto_arima	Total live births	-5255.1	6240.36	5255.1	15.2625	0.569298
naive	Total live births	-5255.1	6240.36	5255.1	15.2625	0.569298

```
knitr::include_graphics(file.path(figure_dir, "holdout_forecast_comparison.png"))
```

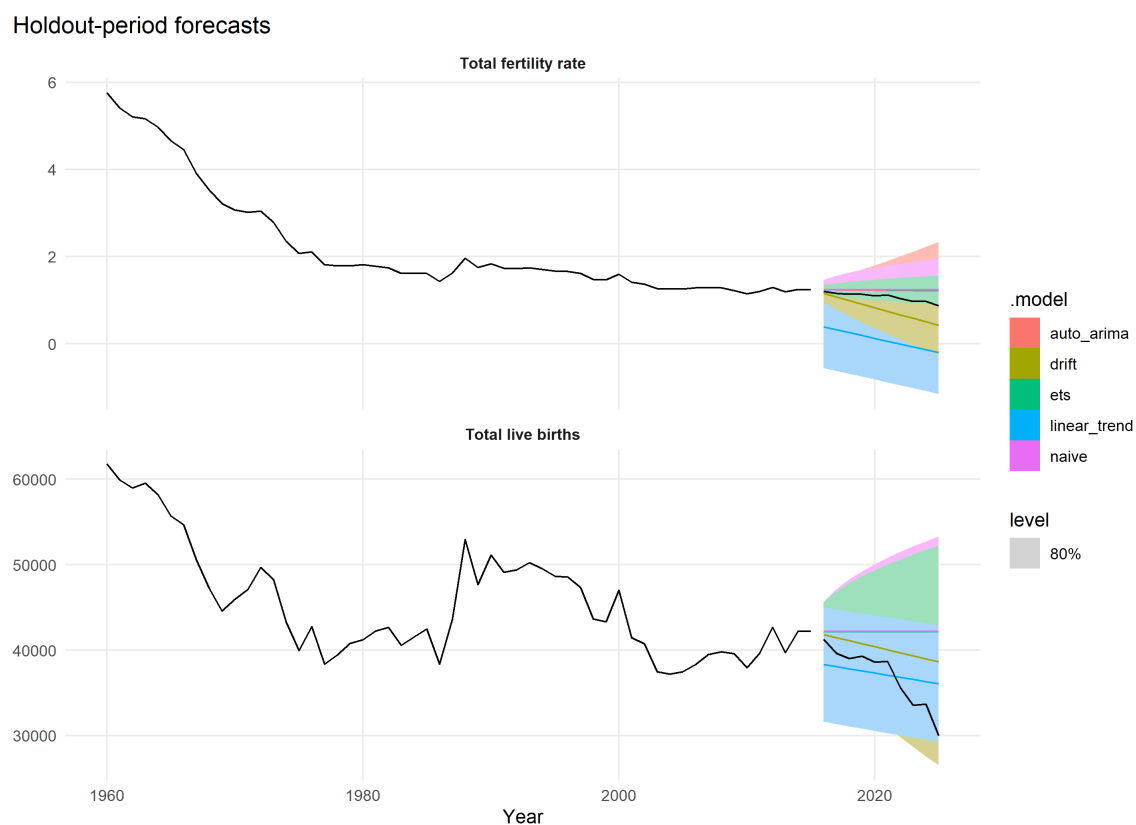


Figure 5: Holdout forecast comparison (2016–2025).

```
best_holdout <- read_csv(file.path(table_dir, "best_holdout_models.csv"),
  show_col_types = FALSE)
kable(best_holdout, digits = 3)
```

Table 9: Best model per series by holdout RMSE (accuracy criterion only)

.model	Measure	.type	ME	RMSE	MAE	MPE	MAPE	MASE	RMSSE	ACF1
auto_arima	Total fertility rate	Test	-	0.178424	0.153068	-	15.2901			0.570258
				0.153068		15.2901				
linear_trend	Total live births	Test	-	2718.99	2321.32	-	6.63675			0.523552
			265.685			1.45324				

Note that `best_holdout_models.csv` identifies the linear trend as best for live births by RMSE alone. The main report selects Drift instead, because the combined accuracy-and-diagnostics criterion is the appropriate basis for model selection (see Section 7.3 of the main report).

7 Reproducibility Notes

```
manifest <- read_csv(file.path(table_dir, "output_manifest.csv"),
                      show_col_types = FALSE)
kable(manifest)
```

Table 10: Generated output manifest

file	output_type	extension	size_bytes	modified_time
outputs/figures/annual_first_differences.figure	figure	png	55613	2026-06-05 05:07:13
outputs/figures/annual_series_overview.figure	figure	png	49978	2026-06-05 05:07:12
outputs/figures/candidate_model_residuals.figure	figure	png	113761	2026-06-05 05:07:16
outputs/figures/candidate_residual_acf.figure	figure	png	81370	2026-06-05 05:07:17
outputs/figures/first_difference_acf.png	figure	png	42608	2026-06-05 05:07:14
outputs/figures/five_year_candidate_forecasts.png	figure	png	62562	2026-06-05 05:07:21
outputs/figures/holdout_forecast_comparison.png	figure	png	60074	2026-06-05 05:07:19
outputs/figures/level_acf.png	figure	png	42117	2026-06-05 05:07:12
outputs/figures/scale_comparison.png	figure	png	53963	2026-06-05 05:07:11
outputs/tables/annual_change_features.csv	table	csv	297	2026-06-05 05:07:11
outputs/tables/annual_first_differences.csv	table	csv	5186	2026-06-05 05:07:14
outputs/tables/best_holdout_models.csv	table	csv	381	2026-06-05 05:07:21
outputs/tables/candidate_model_fit_statistics.csv	table	csv	1734	2026-06-05 05:07:17

file	output_type	extension	size_bytes	modified_time
outputs/tables/candidate_model_registry.csv	table	csv	506	2026-06-05 05:07:17
outputs/tables/candidate_model_terms.csv	table	csv	1305	2026-06-05 05:07:17
outputs/tables/clean_birth_fertility_series.csv	table	csv	4381	2026-06-05 05:07:11
outputs/tables/data_validation_checks.csv	table	csv	417	2026-06-05 05:07:11
outputs/tables/descriptive_features.csv	table	csv	446	2026-06-05 05:07:11
outputs/tables/differenced_series.csv	table	csv	11025	2026-06-05 05:07:11
outputs/tables/forecast_design.csv	table	csv	343	2026-06-05 05:07:17
outputs/tables/holdout_forecast_accuracy.csv	table	csv	1443	2026-06-05 05:07:21
outputs/tables/model_selection_evidence.csv	table	csv	3343	2026-06-05 05:07:21
outputs/tables/output_manifest.csv	table	csv	2815	2026-06-04 01:08:03
outputs/tables/output_quality_checks.csv	table	csv	651	2026-06-04 01:08:03
outputs/tables/report_input_summary.csv	table	csv	335	2026-06-04 01:08:03
outputs/tables/residual_ljung_box_tests.csv	table	csv	691	2026-06-05 05:07:17
outputs/tables/series_association.csv	table	csv	99	2026-06-05 05:07:11
outputs/tables/series_summary.csv	table	csv	185	2026-06-05 05:07:11
outputs/tables/session_info.txt	table	txt	1396	2026-06-04 01:08:03
outputs/tables/stationarity_screen.csv	table	csv	527	2026-06-05 05:07:11
outputs/tables/transformation_recommendation_inputs.csv	table	csv	412	2026-06-05 05:07:11
outputs/tables/transformation_screen.csv	table	csv	1000	2026-06-05 05:07:11
outputs/tables/transformation_series.csv	table	csv	12738	2026-06-05 05:07:11

```
quality_checks <- read_csv(file.path(table_dir, "output_quality_checks.csv"),
                           show_col_types = FALSE)
kable(quality_checks)
```

Table 11: Output quality checks

file	exists	size_bytes	non_empty	check_passed
outputs/tables/data_validation_checks.csv	TRUE	417	TRUE	TRUE
outputs/tables/forecast_design.csv	TRUE	343	TRUE	TRUE
outputs/tables/holdout_forecast_accuracy.csv	TRUE	1443	TRUE	TRUE
outputs/tables/model_selection_evidence.csv	TRUE	3343	TRUE	TRUE
outputs/tables/output_manifest.csv	TRUE	2898	TRUE	TRUE
outputs/figures/annual_series_overview.png	TRUE	49978	TRUE	TRUE
outputs/figures/candidate_residual_acf.png	TRUE	81370	TRUE	TRUE
outputs/figures/holdout_forecast_comparison.png	TRUE	60074	TRUE	TRUE
outputs/figures/five_year_candidate_forecasts.png	TRUE	62562	TRUE	TRUE

```
report_inputs <- read_csv(file.path(table_dir, "report_input_summary.csv"),
                           show_col_types = FALSE)
kable(report_inputs)
```

Table 12: Report input summary

item	status
data_validation	all_validation_checks_passed
required_outputs	required_outputs_present
holdout_window	2016-2025
top_accuracy_model	total_fertility_rate: auto_arima: accuracy_and_residuals
drafting_warning	use_as_planning_input_not_final_prose

The full workflow is reproducible from the project root using:

```
source("scripts/run_project.R")
```

On Windows, the PowerShell helper script is available:

```
.\scripts\run_windows_workflow.ps1
```

R version and package versions at the time of the analysis run recorded above in outputs/tables/session_info.txt. Key packages: fable 0.5.0, feasts 0.5.0, fabletools 0.7.0, tsibble 1.2.0, ggplot2 4.0.3, dplyr 1.2.1. R version 4.4.1 (2024-06-14), running under Windows 11.